CO1-AT3

SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: **Artificial Intelligence** **COURSE CODE: CSA17**

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. *(BL3)*

Assessment Tool Weightage: 10%

QUESTIONS: 2 **Duration : 45 Minutes**
Total Marks:20

Annexure A

Descriptive Written Test

Code of Conduct:

I N.AKhilesh 192365074 *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:
N.Akhilesh Kumar

17/20

Annexure A

Descriptive Written Test

1. A smart home automation system is being designed to manage lighting, temperature, and security based on user behavior and environmental conditions. The system must respond to real-time inputs such as motion detection, temperature changes, and time of day while also learning user preferences over time. In this context, explain the different types of intelligent agents (simple reflex, model-based, goal-based, and utility-based agents) and analyze how each type would function within this system. Justify which type of agent or hybrid approach would be most effective for ensuring comfort, energy efficiency, and security.

2. A robot is placed in an unknown maze and must find a path to the exit without prior knowledge of the layout. Explain how uninformed search strategies like Uniform Cost Search (UCS) and Iterative Deepening Search (IDS) can help solve this problem. Discuss the advantages and disadvantages of these algorithms in terms of time and space complexity. Relate the problem to different types of intelligent agents and explain how agent design affects decision-making. Suggest which combination of agent type and search strategy would be most effective and justify your choice.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Criteria	**Level 4 (Exemplary)**	**Level 3 (Proficient)**	**Level 2 (Developing)**	**Level 1 (Beginning)**	**Max Marks**
Understanding of Concepts	25%	Demonstrates thorough, accurate understanding of concepts	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor understanding; major misconceptions
Explanation	25%	Detailed, well-explained answers with relevant examples	Clear explanation with adequate details	Limited explanation; lacks depth	Poor or unclear explanation
Application	20%	Effectively applies concepts with strong analytical ability	Applies concepts with minor errors	Limited application with weak analysis	No application or incorrect analysis
Organization	15%	Well-structured answers with logical flow and coherence	Organized with minor issues in flow	Basic structure, lacks clarity	Poor organization, incoherent answers
Accuracy	10%	Completely accurate and covers all required points	Mostly accurate with minor omissions	Partially correct with missing points	Incorrect answers with major gaps
Clarity	5%	Clear, neat, professional presentation	Understandable with minor issues	Limited clarity, readability issues	Poorly written, difficult to understand

Smart Home Automation System using Intelligent Agents :-

Introduction

An Intelligent Agent is a system that percives its environment using sensors and performs action using actuators to achieve specific goals.

In Smart Home Automation system, sensors detect:

- Motion
- Temperature
- Light intensity
- Time of day
- Door/window status

The system controls

- Lights
- Air Conditions
- Heater
- Security alarms
- Door locks

1. Simple Reflex Agent

Definition

A Simple Reflex Agent acts only on the current perception using predefined IF-THEN rules.

It does not remember previous events

Working in Smart Home :-

Example rule : * Turn on the light whene there is mothion.
* If temperature is greater than 30°c turn ON AC

Example Rules

If motion detected → Turn On Lights.

If temperature >30°c → Turn ON AC

If doors opens at night → Trigger alarm

2. Model-Based Agent

A model Based Agent maintains an intenal model of the environment and remembers previous states. It are occupied continuously updates its knowledge based on sensor inputs. In a smart home, it can remember which rooms are occupied which lights are already on, and whether doors or windows are open. This memory allows the system to make better decision even when sensors information is incomplete or temporarily unavilable

* Maintains an internal memory.
* Remembers previous states.
* Handles partially observable environments.
* Makes more accurate decisions.
* Better than a Simple Reflex Agent.

3. Goal Based Agent

A Goal-Based Agent makes decisions by considering a specific goal that need to be achieved. Instead of home, the goals may include maintaining a comfortable room temperature reducing electricity consumption, and improve home security.

* Works towards predefined goals.
* Uses planning before taking actions
* Produces better decisions
* More flexible than Reflex agent.

4. Utility Based Agent.

A utility Based Agent selects the actions that provides the highest overall benefits by considering multiple factors such as comfort, energy consumption, security and user satisfaction. Instead of focusing only on one goal, it evaluates the usefulness of every possible action. This enables the system to make balanced decision that satisfy both the user and the environments.

Eg:- Temperature = 27°C

option 1 :-
- AC at 18°C
- High comfort
- High electricity consumption

option 2 :-
- fan + AC at 25°C
- Good comfort
- low electricity consumption

Comparision of Intelligent Agents

Agent Type	Memory	Goal	Decision Method	Smart Home Eg
Simple Reflex	No	No	If-THEN Rules	Motion → ON light
Model based	Yes	No	Internal Model	Remembers occ rooms
Goal Based	Yes	Yes	Planning	Maintain temp
Utility Based	Yes	Yes	Utility function	Balance comfort

2. Robot in an Unknown Maze Using Uniformed Search Startegies

A robot placed inside an unknown maze must search for the exit without having any prior knowledge of the maze structure. This problem can be solved using uninformed search algorithms because the robot has no information about the location of the goal. Two Commonly used uniformed search algorithms are uniform Cost Search [UCS] and Iterative Depending Search [IDS] These algorithms explore different paths until the exit is found.

Example:-

Path A = Cost 5

Path B = Cost 8

Robot selects Path A because it has the lowest cost

Uniform Cost Search:-

Uniform Cost Search is an uniformed search algorithm that always expands the nodes with the lowest cumulative path Cost. Instead of selecting the shortest path based on the number of steps, USC chooses the path with the minimum total cost. Therefore it guarantees finding the optimal solution when all costs are postive.

Iterative Deepening Search

Iterative Deepening Search combines the advantages of Depth first Search and Breadth-first search. It repeatedly performs Depth-Limited Search by increasing the search depth one level at a time util the goal is found. IDS is very useful when the depth of the goal is unnown because it requires much less memory that UCS while still guarenteeing that a solution will eventually be found.

Example :-

Depth 0 → Search Root

Depth 1 → Search Childern

Depth 2 → Search GrandChildern

Continue until Exit is found.

Advantages of DisAdvantages

Uniform Cost Search of Iterative Deepening Search

Uniform Cost Search

* finds the least cost-path
* Complete algorithm
* Optimal solution
* These are Advantages of Uniform Cost Search

Iterative Deepening Search

* Uses very little memory
* Complete algorithm
* Suitable when the goal depth is unknown
* These are the Advantages of the Iterative Deepening Search

Time & Space Complexity

Algorithm	Time Complexity	Space Complexity
Uniform Cost Search	High [depends on path cost]	High
Iterative Deepending Search	$o(b^d)$	$o(b \times d)$

b = branch factor

d = depth.